

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 9: Electronic Flight Instrument Systems (PFD & ND)

Syllabus Source: CAE Oxford (pp. 299-348) | Pass Target: >= 90% AviationExam

1. EFIS ARCHITECTURE: DISPLAY UNITS & SYMBOL GENERATORS

Electronic Flight Instrument Systems (EFIS) replace electromechanical gauges with high-resolution digital screens driven by central graphic processing computers (Display Computers / Symbol Generators - SG):

- Screen Types: Older airliners use Cathode Ray Tubes (CRT). Modern aircraft use Active Matrix Liquid Crystal Displays (AMLCD) providing wider viewing angles, lower power consumption, and daylight readability.
- Display Redundancy & Reversion: Each pilot seat has a Primary Flight Display (PFD) and a Navigation Display (ND). If a PFD screen burns out or fails, an EFIS display transfer switch moves the entire PFD image automatically or manually onto the adjacent ND screen!

2. STANDARD EFIS COLOR CODE PHILOSOPHY

EASA CS-25 certification standardizes EFIS color usage across all manufacturers to prevent pilot misinterpretation:

EFIS Color Standard	Aeronautical Meaning & Cockpit Applications
RED	Warnings, immediate flight hazards, flight envelope red line limits (e.g. V_NE, V_MO / M_MO barber pole, stick shaker speed band, TCAS RA red blocks).
AMBER / YELLOW	Cautions, abnormal conditions, armed limit ranges, airspeed caution range (yellow arc), TCAS TA advisory circles.
GREEN	Engaged active autopilot/flight director modes on the FMA, active navigation tracking (VOR/ILS course line), normal system status.
MAGENTA	Active flight guidance targets! Flight Director steering command bars, active FMS route flight plan, active target altitude, active waypoint name, VNAV descent path.
CYAN (LIGHT BLUE)	Armed autopilot modes on the FMA (waiting for capture), pre-selected bug values, secondary flight routes, tuned navigation aids.
WHITE	Current active numerical flight data, instrument dials and scales, non-active flight plan waypoints, system baseline borders.

3. PRIMARY FLIGHT DISPLAY (PFD) & NAVIGATION DISPLAY (ND) MODES

The PFD centralizes the classic 'Basic T' instruments on a single screen, topped by the Flight Mode Annunciator (FMA):

- Flight Mode Annunciator (FMA): Divided into 5 columns across the top of the PFD: 1. Autothrottle status; 2. Vertical mode (engaged green / armed cyan); 3. Lateral mode (engaged green / armed cyan); 4. Approach capability (CAT I, II, III DUAL); 5. Autopilot / Flight Director engagement status. ANY MODE CHANGE is highlighted by a flashing white box for 10 seconds!
- Speed Trend Vector: A vertical magenta or green trend vector arrow on the airspeed tape. The tip indicates what the airspeed will be in exactly 10 SECONDS based on current longitudinal acceleration!

- ND Modes: 1. ROSE NAV / VOR / ILS: Full 360° compass rose centered on aircraft; 2. ARC MODE: 90° expanded forward sector, track-up oriented, displays active route, weather radar, TCAS and terrain overlays; 3. PLAN MODE: True North-up, centered on selected route waypoints, used for flight plan verification (aircraft symbol is not shown at center).

[!] EXAM TRAP (AVIATIONEXAM) - Navigation Display (ND) Radar Overlay in PLAN Mode

- CRITICAL EXAM RULE: Weather Radar (WXR) returns and Terrain (GPWS) returns CANNOT BE DISPLAYED in PLAN mode! - Reason: PLAN mode is oriented TRUE NORTH UP, whereas the weather radar antenna scans relative to the aircraft heading/track! - Weather radar overlay is available ONLY in ARC mode and ROSE mode.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): EFIS ND PLAN Mode Orientation

Rule 1: ROSE and ARC modes are oriented with aircraft Heading or Track UP and rotate in turns. Rule 2: PLAN mode is ALWAYS oriented with TRUE NORTH UP. It does not rotate. Rule 3: In PLAN mode, center of display is the selected waypoint from FMS route, not the aircraft.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under standard EASA EFIS color coding rules, what color is used on the PFD and ND to represent the active flight director guidance targets, active waypoints, and FMS lateral/vertical profiles.

- [A]: Green
- [B]: Cyan
- [C]: MAGENTA
- [D]: Yellow

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Under CS-25 conventions, MAGENTA is strictly reserved for commanded flight director guidance targets, active waypoints, and FMS lateral/vertical profiles. (Green indicates engaged modes or raw data).

Q2: What does the 'Speed Trend Vector' on a modern PFD airspeed tape indicate?

- [A]: The target cruise speed
- [B]: The predicted airspeed in exactly 10 SECONDS at the current rate of acceleration or deceleration
- [C]: The maximum flap extend speed
- [D]: The windspeed component

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The speed trend vector arrow projects aircraft acceleration. The tip of the vector indicates the airspeed that will be reached in 10 seconds if current acceleration remains constant.

Q3: Why can weather radar (WXR) returns NOT be displayed on the Navigation Display when PLAN mode is selected?

- [A]: Radar transmitter powers off in PLAN mode
- [B]: PLAN mode is oriented True North up, which cannot be synchronized with the heading-referenced radar antenna scan

- [C]: Screen resolution is insufficient
- [D]: TCAS overrides radar

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The weather radar scans relative to the aircraft's physical nose (heading/track). PLAN mode shows a static map oriented True North up, making radar overlay geometrically incompatible.